



INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA  
SALT LAKE

(School of University of Engineering and Management, Kolkata)



Degree: B.Tech. Stream: ALL (Group- A)

Term – I, First Semester Examination, AUGUST - 2025

Subject Code: ESP101A/SDP181A

Subject Name: Essential Studies for Career Development -I/ Competitive  
Aptitude Training -I

Full Marks: 30

Duration: 1 Hour

**Part – I (ESP)**

**Part - A**

**Attempt any 2 questions**

**Each question carries 2 Marks (2×2)**

1. Which line of latitude and longitude divides India into two halves? 2
2. What are substitute goods? Give examples. 2
3. Create a shortest method to evaluate  $\sqrt{1.5625}$  2
4. **Direction: Students need to show the trick** 2  
In a certain code language, 'PICTURE' is written as 'QHDSVQF'. How would 'BROWSER' be written in that same code?  
A. CQVVTDS  
B. CQPVTDS  
C. QPUTDS  
D. CQVPPDS

**Part - B**

**Attempt any 2 questions**

**Each question carries 3 Marks (3×2)**

5. Distinguish between microeconomics and macroeconomics 3

6. **Direction: Students need to show the trick** 3  
 If in a certain code language, 'ITNIETAM' is code for 'INTIMATE', then which of the following words has the code 'TREVNIETARBI'?
- A. INVRETIBRATE  
 B. INVERTIBARTE  
 C. INVERTIBRETA  
 D. INVERTIBRATE
7. Name the border lines of India with the following international countries: 3  
 A. Borderline with Bangladesh  
 B. Borderline with China in the north  
 C. Borderline with China in the east
8. If  $3\sqrt{5} + \sqrt{125} = 17.88$ , then what will be the value of  $\sqrt{80} + 6\sqrt{5}$ ? 3  
 A. 20.15  
 B. 21.25  
 C. 22.35  
 D. 23.45

### Part - C

Attempt any 1 question

Each question carries 5 Marks (5×1)

9. Describe a shortest method to calculate  $(935421 \times 999999)$  5
10. **Direction: Students need to show the trick** 5  
 In a certain code language 'GIVE' is written as 'VIEG' and 'OVER' is written as 'EVRO'. How will 'DISK' be written in that same code?
- A. SIDK  
 B. KISD  
 C. KDSI  
 D. SIKD
11. Explain the concept of 'Unity in Diversity' for India. 5

### Part – II (SDP)

Part A- Descriptive

Attempt any 1 question

Each question carries 5 Marks. (5 × 1)

12. Write a letter to your best friend sharing your experience of your first year in college. 5
13. Write a letter to a childhood friend recalling a funny or touching memory you both shared. 5

**Part B- English Grammar**  
**Attempt any 3 questions**  
**Each question carries 2 Marks. (3 × 2)**

14. In the following question choose the word which is the exact **OPPOSITE** of the given word: 2
- MALICIOUS**
- A. Malevolent
  - B. Spiteful
  - C. Baneful
  - D. Benign
15. In the following the question choose the word which best expresses the **SAME** meaning of the given word: 2
- WISTFUL**
- A. Melancholy
  - B. Scheme
  - C. Hindrance
  - D. Dictator
16. In the following sentence identify if any of the part has an error. Give suitable reasons for your choice: 2
- The sceneries of Kashmir is breathtaking.
- A. The sceneries
  - B. of Kashmir
  - C. is breathtaking
  - D. No error



17. In the following sentence identify if any of the part has an error. Give suitable reasons for your choice:  
 Each student must bring their own notebook.  
 A. Each student  
 B. must bring  
 C. their own notebook

**Part C- Data Interpretation**  
**Attempt any 1 question**  
**Each question carries 4 Marks. (4 × 1)**

18. **Directions to Solve:** (a-d) Study the following table and answer the questions based on it.  
 The following table gives the percentage distribution of population of five states, P, Q, R, S and T on the basis of poverty line and also on the basis of sex.

| State | Percentage of Population below the Poverty Line |       | Proportion of Males and Females |  |
|-------|-------------------------------------------------|-------|---------------------------------|--|
|       | Below Poverty Line                              |       | Above Poverty Line              |  |
|       |                                                 | M : F | M : F                           |  |
| P     | 35                                              | 5 : 6 | 6 : 7                           |  |
| Q     | 25                                              | 3 : 5 | 4 : 5                           |  |
| R     | 24                                              | 1 : 2 | 2 : 3                           |  |
| S     | 19                                              | 3 : 2 | 4 : 3                           |  |
| T     | 15                                              | 5 : 3 | 3 : 2                           |  |

- a. If the male population above poverty line for State R is 1.9 million, then the total population of State R is? 1  
 A. 4.5 million  
 B. 4.85 million  
 C. 5.35 million  
 D. 6.25 million
- b. What will be the number of females above the poverty line in the State S if it is known that the population of State S is 7 million? 1  
 A. 3 million  
 B. 2.43 million  
 C. 1.33 million  
 D. 5.7 million

- c. What will be the male population above poverty line for State P if the female population below poverty line for State P is 2.1 million? 1
- A. 2.1 million  
B. 2.3 million  
C. 2.7 million  
D. 3.3 million
- d. If the population of males below poverty line for State Q is 2.4 million and that for State T is 6 million, then the total populations of States Q and T are in the ratio? 1
- A. 1:3  
B. 2:5  
C. 3:7  
D. 4:9

19. **Directions to Solve:** (a-d) Study the following table and answer the questions based on it.

**Number of Candidates Appeared, Qualified and Selected in a Competitive Examination from Five States Delhi, H.P, U.P, Punjab and Haryana Over the Years 1997 to 2001**

| Year | Delhi |      |      | H.P  |      |      | U.P  |      |      | Punjab |      |     | Haryana |      |     |
|------|-------|------|------|------|------|------|------|------|------|--------|------|-----|---------|------|-----|
|      | App   | Qual | Sel  | App  | Qual | Sel  | App  | Qual | Sel  | App    | Qual | Sel | App     | Qual | Sel |
| 1997 | 8000  | 850  | (94) | 7800 | 810  | (82) | 7500 | 720  | (78) | 8200   |      |     |         |      |     |
|      |       |      | 680  | (85) | 6400 | 700  | (75) |      |      |        |      |     |         |      |     |
| 1998 | 4800  | 500  | (48) | 7500 | 800  | 65   | 5600 | 620  | 85   | 6800   |      |     |         |      |     |
|      |       |      | 600  | (70) | 7100 | 650  | (75) |      |      |        |      |     |         |      |     |
| 1999 | 7500  | 640  | (82) | 7400 | 560  | 70   | 4800 | 400  | 48   | 6500   |      |     |         |      |     |
|      |       |      | 525  | (65) | 5200 | 350  | (55) |      |      |        |      |     |         |      |     |
| 2000 | 9500  | 850  | (90) | 8800 | 920  | 86   | 7000 | 650  | 70   | 7800   |      |     |         |      |     |
|      |       |      | 720  | (84) | 6400 | 540  | (60) |      |      |        |      |     |         |      |     |
| 2001 | 9000  | 800  | (70) | 7200 | 850  | 75   | 8500 | 950  | 80   | 5700   |      |     |         |      |     |
|      |       |      | 485  | (60) | 4500 | 600  | (75) |      |      |        |      |     |         |      |     |

- a. For which state the average number of candidates selected over the years is the maximum? 1
- A. Delhi  
B. H.P  
C. U.P  
D. Punjab

- b. In the year 1997, which state had the lowest percentage of candidates selected over the candidates appeared? 1
- A. Delhi  
B. H.P  
C. U.P
- c. The number of candidates selected from Haryana during the period under review is approximately what percent of the number selected from Delhi during this period? 1
- A. 79.5%  
B. 81%  
C. 84.5%  
D. 88.5%
- d. The percentage of candidates qualified from Punjab over those appeared from Punjab is highest in the year? 1
- A. 1997  
B. 1998  
C. 1999